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Predicting Primary Graft Dysfunction in Systemic Sclerosis Lung Transplantation Using Machine-Learning and CT Features

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Keywords: acute rejection | lung transplantation | machine learning | primary graft dysfunction | scleroderma | systemic sclerosis

ABSTRACT

Introduction: Primary graft dysfunction (PGD) is a significant barrier to survival in lung transplant (LTx) recipients. PGD in patients with systemic sclerosis (SSc) remains especially underrepresented in research.

Methods: We investigated 92 SSc recipients (mean age 51 years ± 10) who underwent bilateral LTx between 2007 and 2020. PGD was defined as grade 3 PGD at 72 h post-LTx. A comprehensive set of CT image features was automatically computed from recipient chest CT scans using deep learning algorithms. Volumetric analysis of recipients' lungs and chest cavity was used to estimate lung-size matching. Four machine learning (ML) algorithms were developed to predict PGD, including multivariate logistic regression, support vector machine (SVM), random forest classifier (RFC), and multilayer perceptron (MLP).

Results: PGD was significantly associated with BMI >30 kg/m^2 ($p = 0.009$), African American race ($p = 0.011$), lower Preop FEV1 ($p = 0.002$) and FVC ($p = 0.004$), longer waitlist time ($p = 0.014$), higher lung allocation score (LAS) ($p = 0.028$), and interstitial lung disease ($p = 0.050$). From CT analysis, PGD was significantly associated with decreased lung volume ($p < 0.001$), increased heart-chest cavity volume ratio ($p < 0.001$), epicardial ($p = 0.033$) and total heart ($p = 0.049$) adipose tissue, and five cardiopulmonary features ($p < 0.050$). Oversized donor allografts estimated using CT analysis were significantly associated with PGD ($p < 0.050$). The MLP model achieved a maximum AUROC of 0.85 (95% CI: 0.81–0.88) in predicting PGD with four features: *Preop FEV1, heart-chest cavity volume ratio, waitlist time, and donor to recipient chest cavity volume ratio*.

Conclusion: CT-derived features are significantly associated with PGD, and models incorporating these features can predict PGD in SSc recipients.

Abbreviations: AUPRC, area under the precision-recall curve; AUROC, area under receiver operating characteristic curve; CI, confidence interval; EAT, epicardial adipose tissue; ILD, interstitial lung disease; LTx, lung transplant; ML, machine learning; MLP, multilayer perceptron; PAH, pulmonary arterial hypertension; PGD, primary graft dysfunction; RFC, random forest classifier; SSc, systemic sclerosis; SVM, support vector machine.

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1 | Introduction

Despite advancements in lung transplant care, primary graft dysfunction (PGD) remains the leading cause of early mortality among lung transplant recipients and significantly limits both short- and long-term survival [1]. PGD affects approximately 10%–30% of patients, with a mortality rate reaching up to 40% among those affected [2]. While the exact pathophysiology of PGD is not fully understood, the predominant theory attributes it to ischemic reperfusion injury (IRI) during transplantation [1, 3]. Established risk factors for PGD encompass donor characteristics (e.g., age, cause of death, smoking history), recipient demographics (e.g., age, female sex, African American race), recipient health factors (e.g., surgical history, pulmonary hypertension, BMI), and intra-operative variables (e.g., lung size mismatch, anastomosis time, ischemic time), and others [2]. Among these risk factors, lung size mismatch occurs in approximately 50% of LTx and is a major contributor to PGD [2, 4, 5].

Systemic sclerosis (SSc) is a connective-tissue disorder characterized by diffuse fibrosis throughout the skin and organs [6]. Pulmonary manifestations, particularly systemic sclerosis-associated interstitial lung disease (SSc-ILD) and pulmonary arterial hypertension (SSc-PAH), represent the leading cause of mortality in patients with SSc [7, 8]. Currently, lung transplantation (LTx) is the only effective treatment in managing end-stage lung disease in this population. However, due to the multi-organ involvement of SSc, LTx remains a rare procedure, accounting for approximately 1% of total LTx procedures [7]. Recent studies have indicated that survival outcomes for SSc are comparable to those of non-SSc recipients, opening the door for a previously contraindicated lung transplantation procedure [7, 9, 10]. However, the pathogenesis of SSc creates unique challenges. For example, patients with SSc often present with lower BMIs, severe esophageal dysmotility, and skin and visceral fibrosis that can restrict lung physiology [10]. These factors introduce a complex interplay of risks for complications such as PGD, which may differ from other LTx populations or even similar autoimmune populations. Mechanisms such as endothelial injury, pro-inflammatory states, and impaired cardiopulmonary reserve inherent to SSc may uniquely influence outcomes such as PGD. As such, continued research into LTx in SSc patients is essential for understanding and predicting major complications, such as PGD, that may be unique to this patient population.

During pre-lung transplant (LTx) evaluations, chest CT scans are routinely used to assess the extent of lung disease. Recent advancements in ML and imaging software have enabled more comprehensive, quantitative analysis of CT images, providing a deeper understanding of anatomical features and disease patterns [11]. Automated CT analysis offers valuable insights that could influence or even predict transplant complications and outcomes. However, the integration of ML and CT-derived features in LTx research remains underexplored.

In a study by Anderson et al. [12], the quantification of adipose tissue on CT scans in LTx recipients revealed a significant association between subcutaneous adipose tissue and an increased risk of PGD ($p = 0.04$). Like many studies on CT body composition, the authors utilize a single-slice cross-sectional area to estimate total adiposity, a potential limiting factor in their analysis. Our literature review also identified only two LTx studies that utilized ML in PGD-related analyses. Stefanuto et al. [13] developed a support vector machine (SVM) model to distinguish patients with grade 3 PGD from those with lower-grade PGD using pulmonary volatile organic compound (VOC) samples. In their study of 35 LTx patients, the SVM model achieved an AUROC of 0.90 after training on 20 VOC features. While the authors employed strategies to limit overfitting, they acknowledged that the small sample size limited the generalizability of their results. Michelson et al. [14] trained multiple ML models with cross-validation on 11 donor and recipient clinical variables, achieving a maximum AUROC of 0.65 in predicting PGD using a K-nearest neighbor's model. The authors of this study did not have access to imaging data. Importantly, none of these previous studies included recipients with a diagnosis of SSc. Additionally, no prior studies have integrated ML methods with comprehensive CT imaging analysis and clinical data, instead relying on either ML or CT data separately.

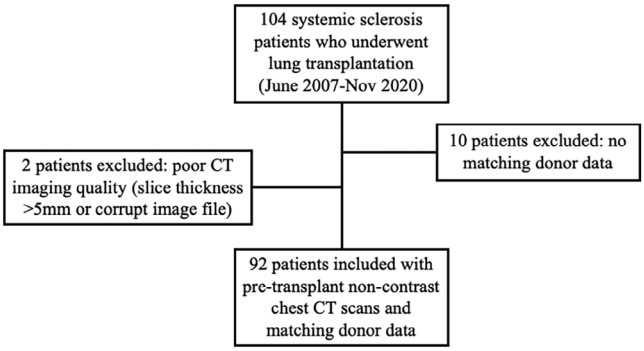

FIGURE 1 | Inclusion flowchart for the study cohort.

We employed deep learning algorithms to automatically segment and quantify lung, chest cavity, cardiopulmonary, and five types of body tissues from standard pre-LTx chest CT scans. Current methods to estimate lung size are based on predictive equations derived from demographics and are largely ineffective [4, 5, 15, 16]. In the case of SSc recipients, we believe equation-based lung size matching may be more inaccurate than in normal recipients due to the fibrotic nature of SSc. Therefore, we also developed a CT-based analysis strategy to estimate size matching through volumetric calculations of recipients' lungs and chest cavities. Four ML algorithms, including multivariate logistic regression, SVM, RFC, and MLP, were trained and evaluated using only pre-LTx clinical and CT-based features to predict PGD occurrence in SSc recipients prior to LTx.

2 | Materials and Methods

2.1 | Study Cohort

We collected a dataset consisting of 92 SSc recipients who underwent bilateral LTx at the University of Pittsburgh Medical Center from 2007 to 2020 (Figure 1). Among these recipients, 21.7% (20/92) developed PGD at 72 h post-operation. The median time between the LTx date and the latest follow-up was 3.02 years (range 4 days to 12.79 years). Pre-LTx non-contrast chest

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CT scans from the patients were acquired using various CT scanners (e.g., Optima CT660 [GE Healthcare], Lightspeed 16 [GE Healthcare], Lightspeed Plus [GE Healthcare], and Lightspeed VCT [GE Healthcare]). Recipients were positioned feet-first and supine for the scans. The scan parameters included a tube voltage of 120 kV and an average tube current of 302 ± 160 mAs. The CT images were reconstructed using the GE "lung" kernel as a matrix of 512 × 512 with an image thickness ranging from 0.625 to 4 mm. The series with the maximum number of CT image slices was selected for the analyses. This study was approved by the University of Pittsburgh Institutional Review Board (IRB #: 24030161).

2.2 | CT Image Features

In-house three-dimensional (3-D) convolutional neural networks (CNN) algorithms [17–19] were used to segment and quantify anatomical structures depicted on recipient chest CT scans, including the lungs, heart, chest cavity, intra- and extra-pulmonary vessels, and body composition tissues. A detailed description of these algorithms and their validation can be found in our publications [17–19]. Briefly, the lung regions were automatically segmented using an adaptive border marching (ABM) computational algorithm, which has been validated extensively in multiple studies involving thousands of CT scans acquired using different protocols and demonstrated promising performance [20–22]. The heart and chest cavity were simply segmented using the classical U-Net model trained and validated on a large collection of CT imaging with manual annotations [15]. Intra- and extra-pulmonary arteries and veins were segmented using our previously developed computer algorithm that combined CNN technology and computational differential geometry to automatically identify pulmonary arteries and veins on non-contrast CT images [17]. This algorithm can identify both extra- and intra-pulmonary arteries and veins. The use of computational differential geometry allows the algorithm to detect very small vessels. Our experiment showed that the algorithm accurately labeled 98% of the intra-pulmonary vessel branches and detected all the branches labeled by radiologists.

Five types of body composition tissues were included: visceral adipose tissue (VAT), subcutaneous adipose tissue (SAT), intermuscular adipose tissue (IMAT), skeletal muscle (SM), and bone. These tissues were segmented and quantified using a 3-D CNN algorithm with a grid-based three-dimensional patch sampling operation [19]. The 3-D CNN model was trained on a cohort of 100 subjects selected from The Cancer Imaging Archive (TCIA), including 50 whole-body CT scans from PET-CT examinations, 25 chest CT scans, and 25 abdominal CT scans. A thoracic radiologist manually outlined VAT, SAT, IMAT, SM, and bones on the CT scans. This algorithm has been validated extensively in multiple studies [20, 23, 24].

Given the identified heart region and body composition, we also segmented and quantified the epicardial adipose tissue (EAT) and the total adipose tissue around the heart. EAT has been implicated in many adverse cardiovascular outcomes and complications [25, 26]; however, no existing research has explored cardiac adipose tissue and lung transplantation outcomes.

Based on these image segmentations, each structure's volume (liters) was computed. We also computed their average densities expressed in Hounsfield Units (HU) for body tissues. Volume ratios between structures were also calculated, such as the heart ratio, calculated by dividing the heart volume (L) by the chest cavity volume (L). These algorithms have been validated across multiple studies [17–19, 27] and perform reliably on various CT protocols and scanners. Due to the unavailability of donor CT imaging, we were unable to generate these metrics from the donors.

2.3 | Clinical Features

From electronic health records (EHR) data, the relevant preoperative, intraoperative, and postoperative variables were collected. Donor characteristics included age, gender, BMI, height, race, and smoking status. Grade 3 PGD was evaluated based on the International Society for Heart and Lung Transplantation (ISHLT) criteria, defined by the presence of pulmonary edema on chest X-ray and a concurrent PaO2/FiO2 ratio < 200 [28]. The presence of ILD or PAH in the SSc recipients was explicitly compared between those who developed PGD and those who did not. Transesophageal echocardiography was used to measure pulmonary artery pressures and evaluate PAH. ILD was evaluated based on a combination of EHR data, pulmonary function tests, CT analysis, and biopsy. Intra-operative transfusion units were calculated as the sum of the number of units of red blood cells, packs of fresh frozen plasma, bags of platelets, and units of cryoprecipitate. Delayed chest closure (DCC) use was recorded in recipients' EHRs for 90 of 92 cases. Graft size mismatch was the leading indication for DCC, representing more than half of the recipients ($n = 18$). Following graft size mismatch, coagulopathy ($n = 14$), reperfusion injury ($n = 4$), and hemodynamics ($n = 2$) were documented as the indications for DCC. Seven cases had multiple indications. All patients underwent a transsternal bilateral thoracotomy incision (clamshell incision) for surgical access. Wedge resection was utilized in 22/92 (23.9%) recipients and lobar resection was used in 8/92 (8.7%) recipients to reduce oversized lung allografts (Table S1).

Due to the lack of direct donor TLC data, the pTLC was estimated using Equation (1), where gender was coded as 0 for female and 1 for male, and race was coded as 0 for non-African American and 1 for African American [5].

$$pTLC = -630.819 + (967.100 * \text{gender}) + (25.197 * \text{height}) - (713.838 * \text{race}) + (15.103 * \text{age}) \quad (1)$$

The pTLC measurements were then used to calculate three ratios for assessing donor-recipient lung size matching. The D_{pTLC}/R_{pTLC} Ratio represents the comparison between the pTLC from donors (D) and recipients (R). The $D_{pTLC}/R_{CT\ Lung\ Vol}$ Ratio compares the donors' pTLC with the recipients' CT-derived lung volume. $D_{pTLC}/R_{CT\ Chest\ Cavity\ Vol}$ Ratio reflects the ratio between the donors' pTLC and the recipients' CT-derived chest cavity volume.

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2.4 | Statistical Analysis

Univariate analysis was performed using a two-tailed Fisher's exact test for categorical variables and the Mann-Whitney U test for continuous variables. All categorical variables were one-hot encoded to enable comparisons. Variables with more than 5% missing data were excluded. The remaining missing data was imputed using the feature mean for continuous variables and the feature mode for categorical variables. A p value of less than 0.05 was considered statistically significant. Given that 173 variables were tested, a summary of the univariate results is provided, with the full analysis available in Table S1.

2.5 | Prediction Modeling

Four state-of-the-art ML models, including SVM, MLP, RFC, and multivariate logistic regression, were chosen to predict the binary outcome of grade 3 PGD at 72 h using pre-LTx features. These models were chosen for their capability to handle multiple input data types and their performance in binary classification tasks. Specifically, SVM and logistic regression models were chosen due to their simplicity and ability to perform robustly on small datasets without overfitting, while RFC and MLP models were selected due to their ability to capture complex, non-linear relationships between features and outcomes [29].

Out of the 173 total variables tested, features with a significance level of $p < 0.10$ were considered for ML analysis to reduce the feature set. Continuous features were z-score standardized. Features with a variance inflation factor (VIF) greater than five were removed to reduce multicollinearity. Lasso regression was then applied to eliminate features with a regression score of zero. Each ML model was trained using repeated stratified 5-fold cross-validation with five runs to ensure robustness and minimize overfitting. During each cross-validation fold, a randomized grid search was performed on the training set to finetune model hyperparameters, including regularization parameters to improve generalization. For this reason, the exact parameters for each ML varied between training sets to optimize performance.

For the RFC model, the hyperparameters explored included 300–500 trees, maximum depth set to none, 10, or 25, and class weights adjusted to either balanced or unbalanced. For the logistic regression model, the regularization strength (C) ranged from 0.1 to 10 with either an L1 or L2 penalty, class weights were balanced or unbalanced, and maximum iterations varied from 500 to 2000. The SVM model hyperparameters featured a regularization strength (C) between 0.1 and 1, a choice between a linear or "rbf" kernel, gamma set to "scaled" or "auto," and class weights as balanced or unbalanced. Last, for the MLP model, the hyperparameters included a single hidden layer with either 50 or 100 neurons, ReLU activation, optimizers Adam or stochastic gradient descent (SGD), L2 regularization alpha ranging from 0.0001 to 0.01, and a maximum number of iterations set to either 500 or 1000. Model performance was evaluated using the area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPRC). All analyses were conducted using scikit-learn 1.5.2 with Python 3.9.1.

3 | Results

3.1 | Significant Clinical Features

Demographic analysis revealed that SSc recipients with a higher BMI ($p = 0.043$), particularly those with a BMI >30 kg/m^2 ($p = 0.009$), as well as African American recipients ($p = 0.011$), experienced significantly higher rates of PGD (Table 1). Recipients exhibited a higher incidence of PGD with decreased pTLC ($p = 0.014$), reduced Preop FEV1 (L) ($p = 0.002$), and reduced Preop FVC (L) ($p = 0.004$) (Table 2). Other factors contributing to increased PGD rates included longer waitlist times ($p = 0.014$), a higher D_{pTLC}/R_{pTLC} Ratio ($p = 0.016$), and a higher LAS ($p = 0.028$). Moreover, recipients diagnosed with SSc-ILD ($p = 0.050$) were more likely to develop PGD than those with SSc-PAH ($p = 0.050$) (Table 2). Intra-operatively, recipients who underwent delayed chest closure were significantly more likely to experience PGD ($p = 0.001$). Lung size mismatch as the indication for DCC was more significantly associated with PGD (40%) recipients over non-PGD (13.9%) recipients ($p = 0.021$). Post-operatively, PGD was associated with the need for mechanical support ($p < 0.001$), longer total vent duration ($p = 0.004$), and increased mortality during the index admission ($p = 0.038$) (Table 2).

3.2 | Significant CT-Features

Seven measurements related to lung size and donor-recipient lung size matching showed a significant association with PGD occurrence. Both recipient lung size, namely recipient pTLC ($p = 0.014$) and CT lung volume ($p < 0.001$), were lower in recipients who developed PGD compared to those who did not (Table 3). A smaller CT chest cavity volume was also linked to PGD ($p < 0.001$) (Table 3). All three ratios estimating lung size matching were also significantly associated with PGD (Figure S1). A larger $D_{pTLC}/R_{CT\ Lung\ Vol}$ Ratio suggested a greater mismatch or larger donor allograft ($p < 0.001$) (Table 3). Similarly, both a larger D_{pTLC}/R_{pTLC} Ratio ($p = 0.016$) and a larger $D_{pTLC}/R_{CT\ Chest\ Cavity\ Vol}$ Ratio ($p < 0.001$) were associated with PGD, potentially indicating oversized lung allografts contributed to PGD development. Furthermore, a lower $R_{CT\ Lung\ Vol}/R_{CT\ Chest\ Cavity\ Vol}$ was associated with PGD ($p = 0.012$), which may be due to increased fibrosis of lung tissue, leading to a reduced lung-to-chest cavity ratio and subsequent lung mismatching. Notably, donor pTLC was the only lung measurement that did not differ significantly between PGD and non-PGD groups ($p = 0.917$) (Table 3).

In the univariate analysis, three heart-related recipient features and five cardiopulmonary features were significantly associated with PGD. A larger heart-chest cavity volume ratio ($p < 0.001$) was associated with PGD, potentially indicating the presence of cardiomegaly as this ratio reflects an enlarged heart volume relative to a recipient's chest cavity size (Table 3). Additionally, higher densities of epicardial adipose tissue ($p = 0.033$) and total heart adipose tissue density ($p = 0.050$) were associated with PGD. Lower intrapulmonary vein volume ($p = 0.002$) and vein-chest cavity volume ratio ($p = 0.050$) were also associated with PGD.

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TABLE 1 | Recipient and donor characteristics ($n = 92$).

Characteristic	Statistic	PGD ($n = 20$)	Non-PGD ($n = 72$)	p value
Age (years)				
Mean ± SD	51 ± 10	50 ± 9	51 ± 10	0.705
Gender				
Female	56/92 (61.0%)	15/20 (75.0%)	41/72 (56.9%)	
Male	36/92 (39.0%)	5/20 (25.0%)	31/72 (43.1%)	0.197
BMI (kg/m^2)				
Mean ± SD	25.1 ± 4.9	27.2 ± 5.8	24.5 ± 4.5	**0.043**
<18.5	7/92 (7.6%)	1/20 (5.0%)	6/72 (8.3%)	1.000
18.5–24.9	40/92 (43.5%)	6/20 (30.0%)	34/72 (47.2%)	0.208
25.0–29.9	28/92 (30.4%)	5/20 (25.0%)	23/72 (31.9%)	0.784
>30.0	17/92 (18.5%)	8/20 (40.0%)	9/72 (12.5%)	**0.009**
Height (cm)				
Mean ± SD	168.4 ± 9.1	167.0 ± 9.4	168.8 ± 9.1	0.628
Race				
African American	20/92 (21.7%)	9/20 (45.0%)	11/72 (15.3%)	**0.011**
White	68/92 (73.9%)	10/20 (50.0%)	58/72 (80.6%)	**0.010**
Smoking status				
Former	39/92 (42.4%)	7/20 (35.0%)	32/72 (44.4%)	0.610
>20 pack-years	17/92 (18.5%)	2/20 (10.0%)	15/72 (20.8%)	0.346
Donor age (years)				
Mean ± SD	36 ± 15	33 ± 11	36 ± 16	0.391
Donor gender				
Female	61/92 (66.3%)	13/20 (65.0%)	48/72 (66.7%)	
Male	31/92 (33.7%)	7/20 (35.0%)	24/72 (33.3%)	1.000
Donor BMI (kg/m^2)				
Mean ± SD	26.2 ± 5.7	27.0 ± 7.0	26.0 ± 5.4	0.824
Donor height (cm)				
Mean ± SD	164.8 ± 9.8	165.8 ± 7.9	164.5 ± 10.3	0.935
Donor race				
African American	23/92 (25.0%)	3/20 (15.0%)	20/72 (27.8%)	0.382
White	61/92 (66.3%)	16/20 (80.0%)	45/72 (62.5%)	0.186
Donor smoking status				
>20 pack-years	11/92 (12.0%)	3/20 (15.0%)	8/72 (11.1%)	0.699

Abbreviations: PGD, primary graft dysfunction; pTLC, predicted total lung capacity (based on age, gender, race, and height); SSc, systemic sclerosis; SD, standard deviation.

Furthermore, lower intrapulmonary artery volume ($p = 0.023$), higher artery-vein volume ratio ($p = 0.050$), and higher artery-chest cavity volume ratio were each independently associated with PGD. Collectively, these findings suggest a relationship between the baseline cardiopulmonary pathophysiology of SSc recipients and the subsequent development of PGD following LTx (Table 3).

None of the five body composition tissue measurements showed a significant association with PGD development, as detailed in Table S1.

3.3 | ML Prediction Models

After applying VIF and Lasso regression feature elimination, four key pre-LTx features were selected, including *heart-chest cavity volume ratio, waitlist time,* $D_{pTLC}/R_{CT\ Chest\ Cavity\ Vol}$ *Ratio*, and *Preop: FEV1*. This feature set was used by all ML models to retrospectively predict the occurrence of PGD. Using this feature set, the RFC model achieved an AUROC of 0.77 (0.73–0.82) and an AUPRC of 0.51 (0.42–0.60). The SVM model performed better, with an AUROC of 0.83 (95% CI: 0.79–0.86) and an AUPRC of 0.58 (95% CI: 0.49–0.66). The MLP model further improved these

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TABLE 2 | Univariate association of clinical features with PGD including post-operative outcomes ($n = 92$).

Characteristic	Statistic	PGD ($n = 20$)	Non-PGD ($n = 72$)	p value
Preoperative features				
Preop: FEV1 (L)	1.39 ± 0.53	1.08 ± 0.34	1.48 ± 0.55	**0.002**
Preop: FVC (L)	1.78 ± 0.71	1.38 ± 0.42	1.89 ± 0.74	**0.004**
Waitlist Time (days)	162 ± 198	283 ± 241	128 ± 171	**0.014**
D_{pTLC}/R_{pTLC} Ratio	0.93 ± 0.20	1.03 ± 0.21	0.91 ± 0.18	**0.016**
LAS	61.6 ± 18.8	68.9 ± 18.5	59.5 ± 18.5	**0.028**
SSc-PAH	27/92 (29.3%)	2/20 (10.0%)	25/72 (34.7%)	**0.050**
SSc-ILD	65/92 (70.7%)	18/20 (90.0%)	47/72 (65.3%)	**0.050**
ACE inhibitor therapy	13/92 (14.1%)	0/20 (0.0%)	13/72 (18.1%)	0.064
Tracheostomy	42/92 (45.7%)	13/20 (65.0%)	29/72 (40.3%)	0.075
Ischemic Reperfusion Injury	60/92 (65.2%)	10/20 (50.0%)	50/72 (69.4%)	0.119
Previous Chest Operation	21/92 (22.8%)	5/20 (25.0%)	16/72 (22.2%)	0.770
Intra-operative features				
Delayed Chest Closure (DCC)	32/92 (34.8%)	14/20 (70.0%)	18/72 (25.0%)	**0.001**
DCC for Size Mismatch	18/92 (19.6%)	8/20 (40.0%)	10/72 (13.9%)	**0.021**
CPB Support	43/92 (46.7%)	12/20 (60.0%)	31/72 (43.1%)	0.211
Transfusion Units	13.1 ± 22.0	10.4 ± 9.1	13.8 ± 24.4	0.453
VA-ECMO Support	29/92 (31.5%)	7/20 (35.0%)	22/72 (30.6%)	0.787
Washout	14/92 (15.2%)	3/20 (15.0%)	11/72 (15.3%)	1.000
ECMO Bridge to Transplant	9/92 (9.8%)	2/20 (10.0%)	7/72 (9.7%)	1.000
MV Bridge to Transplant	11/92 (12.0%)	2/20 (10.0%)	9/72 (12.5%)	1.000
Postoperative outcomes				
Postop Mechanical Support	25/92 (27.2%)	15/20 (75.0%)	10/72 (13.9%)	**<0.001**
Total Vent Duration (Hours)	422 ± 762	507 ± 606	399 ± 802	**0.004**
Mortality on Index Admission	7/92 (7.6%)	4/20 (20.0%)	3/72 (4.2%)	**0.038**
Acute Kidney Injury	38/92 (41.3%)	12/20 (60.0%)	26/72 (36.1%)	0.073
ICU LOS	20.8 ± 31.7	21.1 ± 19.1	20.7 ± 34.5	0.088
30-Day Mortality	2/92 (2.2%)	2/20 (10.0%)	0/72 (0.0%)	**<0.001**
90-Day Mortality	7/92 (7.6%)	5/20 (25.0%)	3/72 (4.2%)	**0.029**
1-Year Mortality	22/92 (23.9%)	7/20 (35.0%)	15/60 (20.0%)	0.277
5-Year Mortality	41/92 (44.6%)	7/20 (35.0%)	33/60 (45.8%)	0.384

Abbreviations: CT Vol, CT-derived lung volume; D, donor; DCC, delayed chest closure; ILD, interstitial lung disease; LAS, lung allocation score; LOS, length of stay; LTx, lung transplant; MV, mechanical ventilation; PAH, pulmonary arterial hypertension; PGD, primary graft dysfunction; pTLC, predicted total lung capacity; R, recipient; SSc, systemic sclerosis; VAECMO, veno-arterial ECMO.

metrics, achieving an AUROC of 0.85 (95% CI: 0.81–0.88) and an AUPRC of 0.61 (95% CI: 0.53–0.68). The multivariate logistic regression model also performed well, with an AUROC of 0.84 (95% CI: 0.81–0.88) and the highest AUPRC among all models at 0.62 (95% CI: 0.54–0.70). Notably, the MLP model exhibited the highest AUROC, while the Logistic Regression model had the best AUPRC (Table 4; Figure 2).

4 | Discussion

PGD continues to be a significant barrier to improving the survival rate among lung transplant recipients. Previous studies in PGD prediction primarily relied on clinical variables [1–3] only and did not include recipients with SSc. Few investigations have utilized ML models or CT imaging features to better understand and predict PGD in LTx recipients. To our knowledge, our study is the first to incorporate both a comprehensive set of CT imaging features and clinical variables using ML models for PGD prediction. In a related study, Stefanuto et al. [13] achieved an impressive AUROC of 0.90 using an SVM trained on VOC samples, but they did not utilize pre-LTx features and worked with a smaller cohort ($n = 35$). Michelson et al. [14] developed and evaluated ML models (AUC = 0.65) using pre-LTx features to predict PGD but relied solely on traditional clinical features. Additionally, Anderson et al. [12] identified an association between subcutaneous adipose

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TABLE 3 | Univariate association of CT-derived thoracic features with PGD ($n = 92$).

Characteristic	Statistic	PGD ($n = 20$)	Non-PGD ($n = 72$)	p value
Lung size and allograft matching features				
$D_{pTLC}/R_{CT\ Lung\ Vol}$ Ratio	1.55 ± 0.56	1.93 ± 0.44	1.45 ± 0.55	**0.0001**
CT Lung Vol (L)	3.01 ± 1.03	2.31 ± 0.51	3.20 ± 1.06	**0.0002**
$D_{pTLC}/R_{CT\ Chest\ Cavity\ Vol}$ Ratio	0.95 ± 0.29	1.13 ± 0.26	0.89 ± 0.28	**0.0003**
CT Chest Cavity Vol (L)	4.78 ± 1.38	3.94 ± 0.87	5.02 ± 1.40	**0.0005**
$R_{CT\ Lung\ Vol}/R_{CT\ Chest\ Cavity\ Vol}$	0.62 ± 0.08	0.59 ± 0.06	0.63 ± 0.08	**0.0116**
R_{pTLC}	4.6 ± 0.8	4.3 ± 0.8	4.7 ± 0.8	**0.0141**
D_{pTLC}/R_{pTLC} Ratio	0.93 ± 0.20	1.03 ± 0.21	0.91 ± 0.18	**0.0164**
D_{pTLC}	4.2 ± 0.7	4.3 ± 0.6	4.2 ± 0.7	0.9174
Recipient heart features				
Heart-Chest Cavity Vol. Ratio	0.14 ± 0.04	0.17 ± 0.03	0.14 ± 0.04	**0.0003**
Epicardial Adipose Tissue Density (HU)	−80.9 ± 10.2	−77.3 ± 7.1	−81.9 ± 10.8	**0.0328**
Total Heart Adipose Tissue Density (HU)	−66.2 ± 14.4	−61.4 ± 9.9	−67.6 ± 15.2	**0.0495**
Heart Vol. (mL)	680 ± 258	676 ± 173	682 ± 278	0.6805
Recipient cardiopulmonary features				
Vein Vol (mL)	54.4 ± 29.6	38.2 ± 21.2	58.9 ± 30.1	**0.0017**
Artery Volume (mL)	79.7 ± 39.5	62.7 ± 27.1	84.4 ± 41.3	**0.0228**
Vein-Chest Cavity Vol. Ratio	0.41 ± 0.08	0.37 ± 0.07	0.42 ± 0.09	**0.0495**
Artery-Vein Vol. Ratio	1.56 ± 0.52	1.81 ± 0.58	1.50 ± 0.49	**0.0495**
Artery-Chest Cavity Vol. Ratio	0.59 ± 0.08	0.63 ± 0.07	0.58 ± 0.09	**0.0495**

Abbreviations: CT Vol, CT-derived lung volume; D, donor; LTx, lung transplant; PGD, primary graft dysfunction; pTLC, predicted lung capacity; R, recipient; SSc, systemic sclerosis; Vol., volume.

TABLE 4 | Performance of machine-learning models in PGD prediction ($n = 92$)

Model	AUROC	AUROC 95% CI	AUPRC	AUPRC 95% CI
RFC	0.77	(0.73–0.82)	0.51	(0.42–0.60)
SVM	0.83	(0.79–0.86)	0.58	(0.49–0.66)
MLP	**0.85**	(0.81–0.88)	0.61	(0.53–0.68)
Multivariate LR	0.84	(0.81–0.88)	**0.62**	(0.54–0.70)

Abbreviations: LR, logistic regression; MLP, multilayer perceptron; PGD, primary graft dysfunction; RFC, random forest classifier; SVM, support vector machine.

tissue and PGD development in a large, multicenter study on body composition. We did not observe the same association, which may stem from differences in patient cohorts and their use of single-slice CT measurements, while we used 3D-CNN quantification. Notably, none of the mentioned studies evaluated donor-recipient lung size matching, a significant contributor to PGD. As our results revealed, recipients who developed PGD were more likely to have lung size mismatching requiring delayed chest closure. Overall, our study explores PGD in an understudied patient population through multiple approaches, integrating ML models, CT imaging analysis, CT-based lung size matching, and clinical variables. The ML models developed in this study demonstrated promising performance, with AUCs ranging from 0.77 to 0.85. Notably, these models demonstrated an ability to predict PGD pre-LTx using only four features. We believe that further investigation of these ML models with CT and clinical features could pave the way for practical risk-assessment applications. This initial study provides evidence that pre-LTx screening for SSc patients at risk of developing PGD is possible. Using AI-based metrics, these patients could be identified and, therefore, benefit from closer perioperative monitoring and more targeted interventions.

We observed that certain clinical features significantly increased the risk of PGD occurrence. Specifically, African American recipients and those with a high BMI were associated with a higher incidence of PGD, consistent with findings from other PGD studies [1–3, 12, 14]. Notably, the ILD phenotype of SSc was found to be more strongly associated with PGD than the PAH phenotype ($p = 0.05$). Furthermore, delayed chest closure ($p = 0.001$) in this cohort was associated with PGD, an intriguing finding since delayed chest closure is often used as an intra-operative strategy

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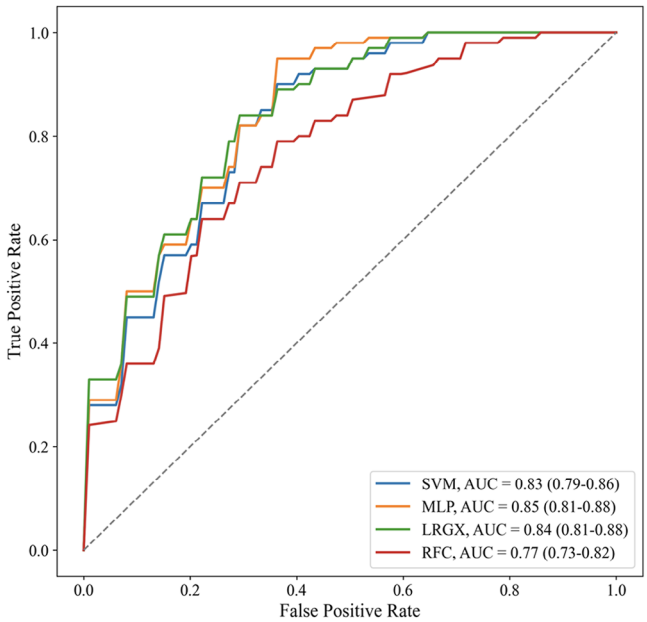

FIGURE 2 | Performances of ML models on PGD prediction using four features: waitlist time, preop FEV1 (L), recipient heart-to-chest cavity volume ratio, donor pTLC-to-recipient chest cavity volume ratio.

to minimize PGD occurrence and other complications [30, 31]. This result likely reflects the fact that DCC was more frequently utilized in severe, complex cases. While prior studies on LTx in SSc recipients have primarily focused on comparing survival and complications between SSc and non-SSc recipients [7–10, 32], our study took a different approach by investigating the clinical and imaging features associated with PGD, aiming to provide clinicians with more specific insights into the SSc-specific risk factors for PGD.

Multiple studies have established that lung allograft size mismatch increases the risk of PGD and post-LTx mortality [2, 4, 16]. Currently, pTLC estimations are largely employed to match recipients to donor lungs; however, these estimates may overstate lung size, particularly in cases of fibrosis-related lung disease. In contrast, CT-based volumetric analysis offers individualized and precise lung size measurements that may be superior to these equations. Our findings indicate that lower recipient lung volumes, as measured by pTLC ($p = 0.014$) and CT ($p < 0.001$), are associated with PGD. Furthermore, the three ratios used to directly assess donor-lung size matching showed significant differences between PGD and non-PGD groups, highlighting that both large mismatches and oversized donor grafts were associated with PGD in SSc recipients. Interestingly, previous studies [4, 16, 33] have reported that oversized donor grafts may reduce the risk of PGD, a finding that contrasts with our results. It is important to note that LTx in SSc is rarely represented in such studies due to its low incidence. Clinically, patients with SSc often physiologically present with smaller, more restricted chest cavities and reduced thoracic compliance due to the fibrotic nature of the disease. In such cases, oversized lung grafts may not conform well to the recipient's thoracic space, potentially increasing mechanical stress, impairing graft expansion, and contributing to allograft injury, ultimately raising the risk of PGD. We acknowledge that future research is needed to validate this claim.

Our study also uniquely measured chest cavity volume to estimate lung-size matching, revealing that both smaller chest cavities and oversized donor grafts significantly contribute to PGD occurrence. Additionally, the recipient's lung-to-chest cavity volume ratio was significant between PGD and non-PGD groups ($p = 0.012$), indicating that the fibrotic nature of lungs in SSc may contribute to PGD and size mismatches. By evaluating donor-recipient size matching using differing modalities (CT vs. pTLC) and from different angles (lungs vs. chest cavity), we identified multiple features that can aid in estimating donor-recipient lung size matching and predicting PGD risk. To our knowledge, this study is also the first to examine the association between cardiopulmonary and heart-related CT features and PGD. These findings warrant further investigation, but they suggest that a recipient's underlying cardiovascular health may be a risk factor for PGD. Importantly, all of these features are automatically extracted from standard CT imaging used in LTx, opening the door for these algorithms to improve clinical decision-making at the time of scan.

This study has several limitations. First, the sample size ($n = 92$) was small and derived from a single center. We fully acknowledge the importance of external validation in strengthening the generalizability of our findings. However, we would like to highlight the inherent challenges in assembling an external cohort for this specific population. Of note, this dataset represents the largest cohort of LTx patients with SSc who have undergone CT scans to date. The rarity of centers performing LTx in SSc patients, combined with the low incidence of the disease, makes it challenging to find an external validation cohort. To mitigate the concern related to the size of our dataset and the risk of overfitting in our models, we employed multiple strategies. We utilized repeated stratified cross-validation to enhance the robustness of our findings. Additionally, we limited our models' input to only four features, achieving a feature-sample ratio of 23:1, further reducing the potential for overfitting. Also, the consistent performance across all four ML models strengthens our confidence in the results. After 2021, only six SSc cases were identified in our database; however, only two had the necessary matching donor data for our analysis. Due to this extremely small sample size ($N = 2$), we did not include these cases in the current analysis or use them as a validation cohort. Second, donor CT imaging was not available for volumetric analysis of donor lungs. Instead, we relied on established pTLC equations to analyze donor-recipient lung ratios. We hope to collect donor CT imaging to validate our results in future studies; however, our initial results show that estimating lung size mismatch can be augmented using CT lung and chest cavity volumes. Third, the CT scans were performed over 13 years using different protocols and scanners, which introduced variations in image quality. However, our 3D-CNN algorithms have been developed and extensively validated against a wide range of CT scans acquired using various protocols, which minimizes the impact of data heterogeneity. In the future, we aim to acquire larger datasets with external validation to further validate our findings. By finetuning our models on these datasets, we hope to conduct prospective trials to assess the impact of our AI systems on patient outcomes for this rare disease.

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5 | Conclusion

As lung transplant centers increasingly consider LTx for patients with SSc, it becomes crucial to thoroughly investigate the factors associated with mortality in this population. Our study contributes valuable insights into the preoperative risk factors for PGD, a significant barrier to survival in these patients. We identified multiple risk factors, including CT-based features that assess lung size matching and indicators of cardiovascular health. The ML models we developed were robust against overfitting and demonstrated an impressive ability to differentiate between recipients likely to develop PGD based solely on pre-LTx data. This capability underscores the potential of these models to be integrated into clinical workflows, enabling the generation of individualized PGD risk scores for prospective recipients. Such risk stratification could significantly enhance clinical decision-making, allowing for more tailored patient management and optimizing outcomes for SSc patients undergoing LTx.

Author Contributions

Jatin Singh participated in the writing of the paper, the performance of the research, and data analysis. Xin Meng participated in the writing of the paper and data analysis. Joseph K. Leader participated in the writing of the paper. John Ryan participated in the writing of the paper. Rachel Deitz participated in the performance of the research and data collection. Lucas Pu participated in the data analysis. Ernest G. Chan participated in data collection and data analysis. Norihisa Shigemura participated in the writing of the paper. Chadi A. Hage participated in the performance of the research and data collection. Pablo G. Sanchez participated in the performance of the research and data collection. Jiantao Pu participated in the writing of the paper, performance of the research, and data analysis.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supplemental Figure S1. Boxplot comparing CT-based donor-recipient lung-size matching ratios and PGD. **Supplemental Table S1.** Complete univariate association of all CT-derived features with PGD (n-92)

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